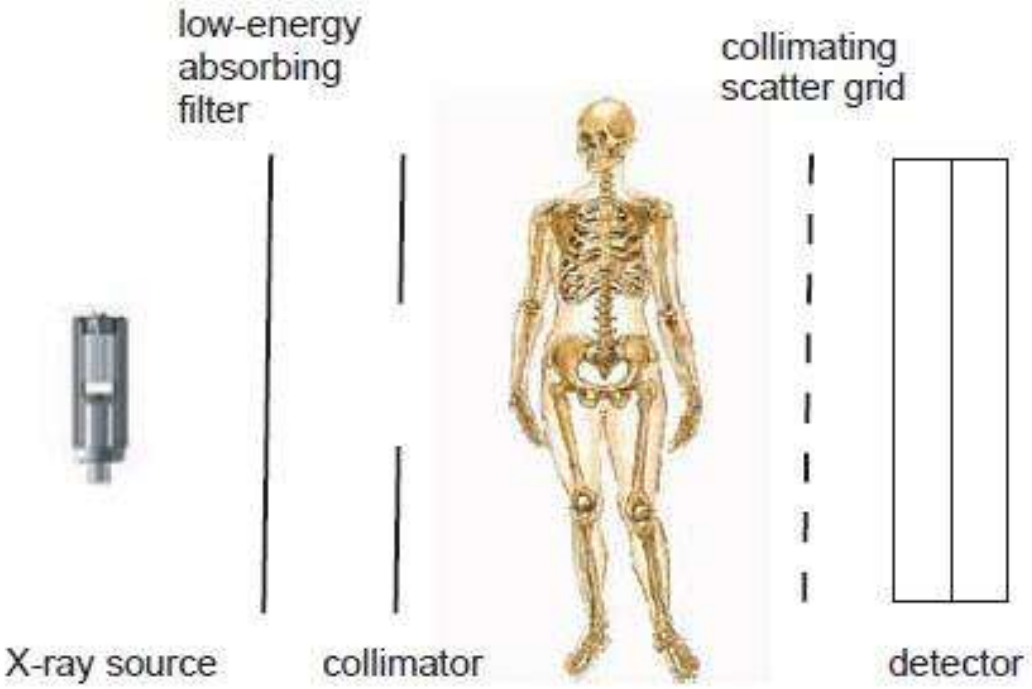
Introdução e Fundamentos

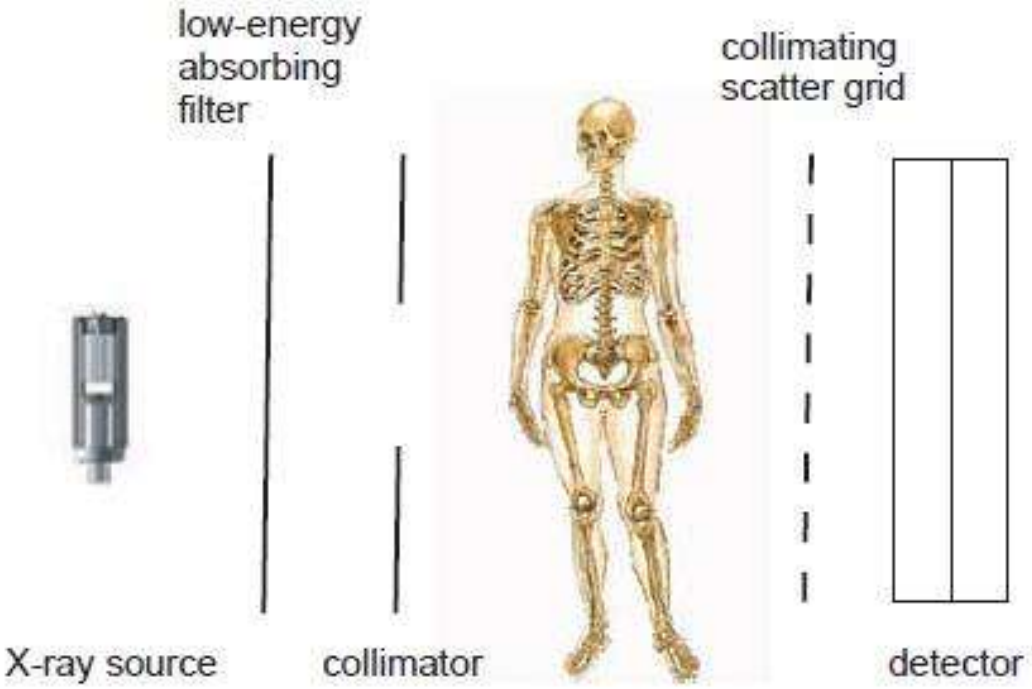
- Aplicações médicas - radiografia



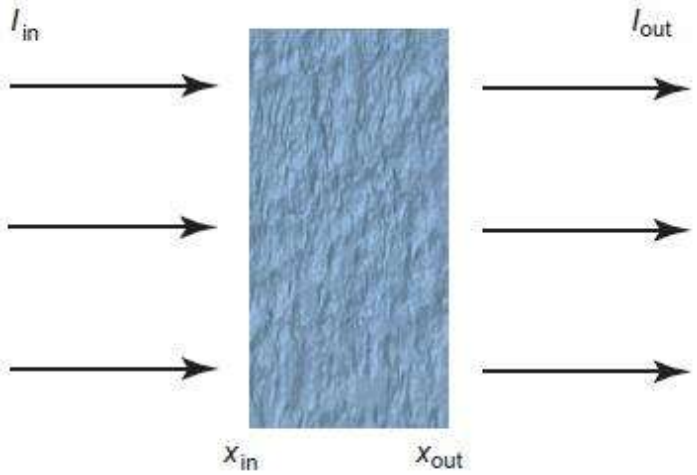
Schematic representation of the radiographic imaging chain.

Introdução e Fundamentos

- Aplicações médicas - radiografia



Schematic representation of the radiographic imaging chain.



$$I_{out} = I_{in}\, e^{-\int_{x_{in}}^{x_{out}} \mu(x)\, dx}$$

Introdução e Fundamentos

- Aplicações médicas - radiografia

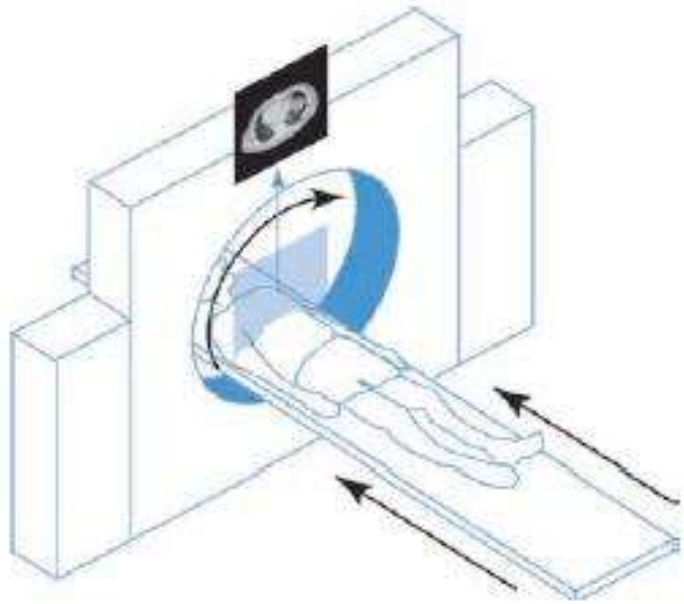


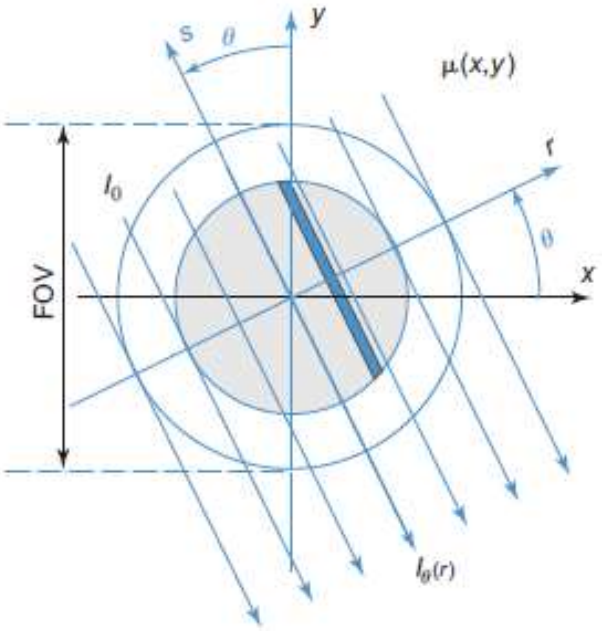
(a) (b)

(a) Dense opacity with spicular border in the cranial part of the right breast; histological proven invasive ductal carcinoma.
(b) Cluster of irregular microcalcifications suggesting a low differentiated carcinoma. (Courtesy of Dr. Van Ongeval, Department. of Radiology.)

Introdução e Fundamentos

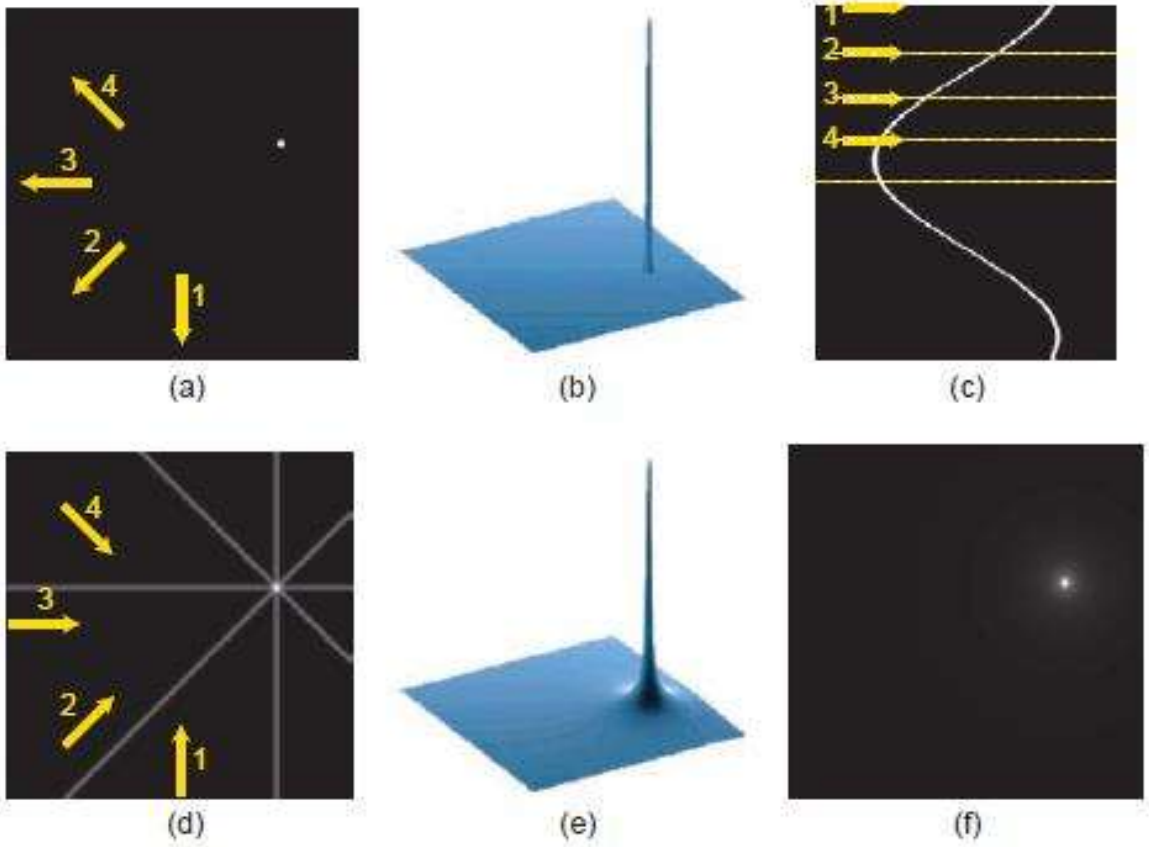
- Aplicações médicas - CT





Introdução e Fundamentos

- Aplicações médicas - CT

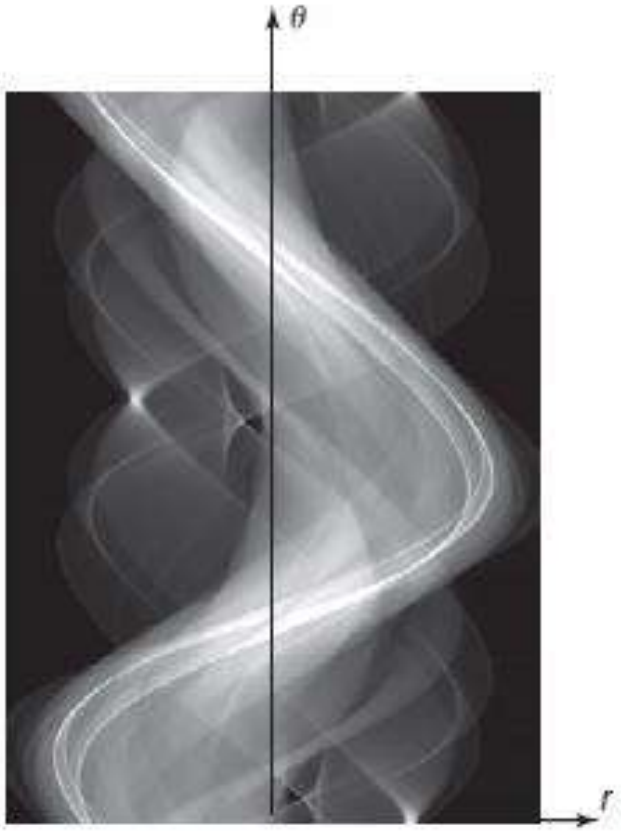


(a and **b)** Image and surface plot of a distribution $\mu(x, y)$ containing one single dot. The arrows indicate four arbitrary projection directions. **(c)** 360°-sinogram obtained by projecting $\mu(x, y)$. The arrows indicate the views that correspond to the four projection directions in (a).

(d) Backprojection of the four views chosen in (a). **(e** and **f)** Surface plot and image of the straightforward backprojection of the entire sinogram in (c).

Introdução e Fundamentos

- Aplicações médicas – CT (sinograma)



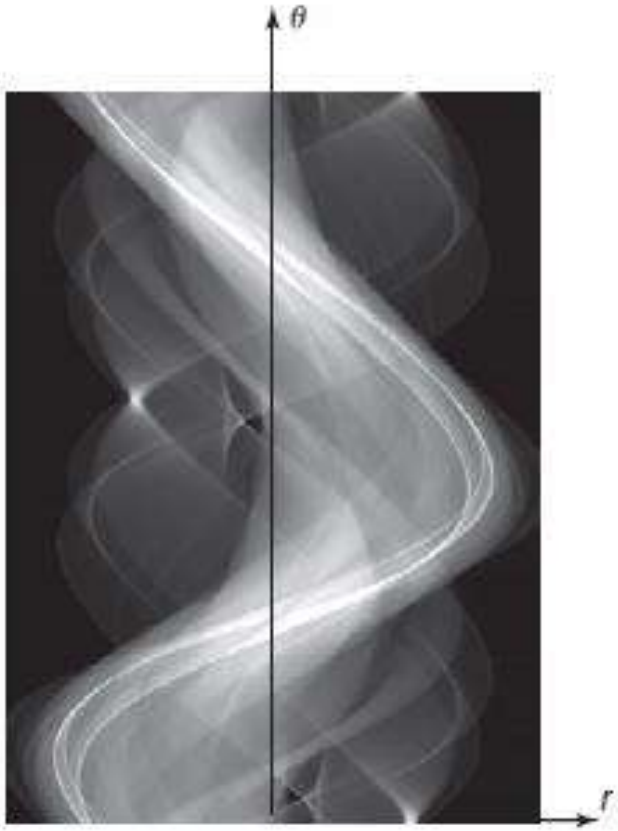
A sinogram is a 2D dataset $p(r,\theta)$ obtained by stacking the 1D projections $p_\theta(r)$.

In mathematics, the transformation of any function $f(x,y)$ into its sinogram $p(r,\theta)$ is called the *Radon transform*:

$$p(r,\theta) = \mathcal{R}\{f(x,y)\} = \int_{-\infty}^{\infty} f(r\cdot\cos\theta - s\cdot\sin\theta, r\cdot\sin\theta + s\cdot\cos\theta)\,\mathrm{d}s.$$

Introdução e Fundamentos

- Aplicações médicas – CT (sinograma)



A sinogram is a 2D dataset $p(r,\theta)$ obtained by stacking the 1D projections $p_\theta(r)$.

In mathematics, the transformation of any function $f(x,y)$ into its sinogram $p(r,\theta)$ is called the *Radon transform*:

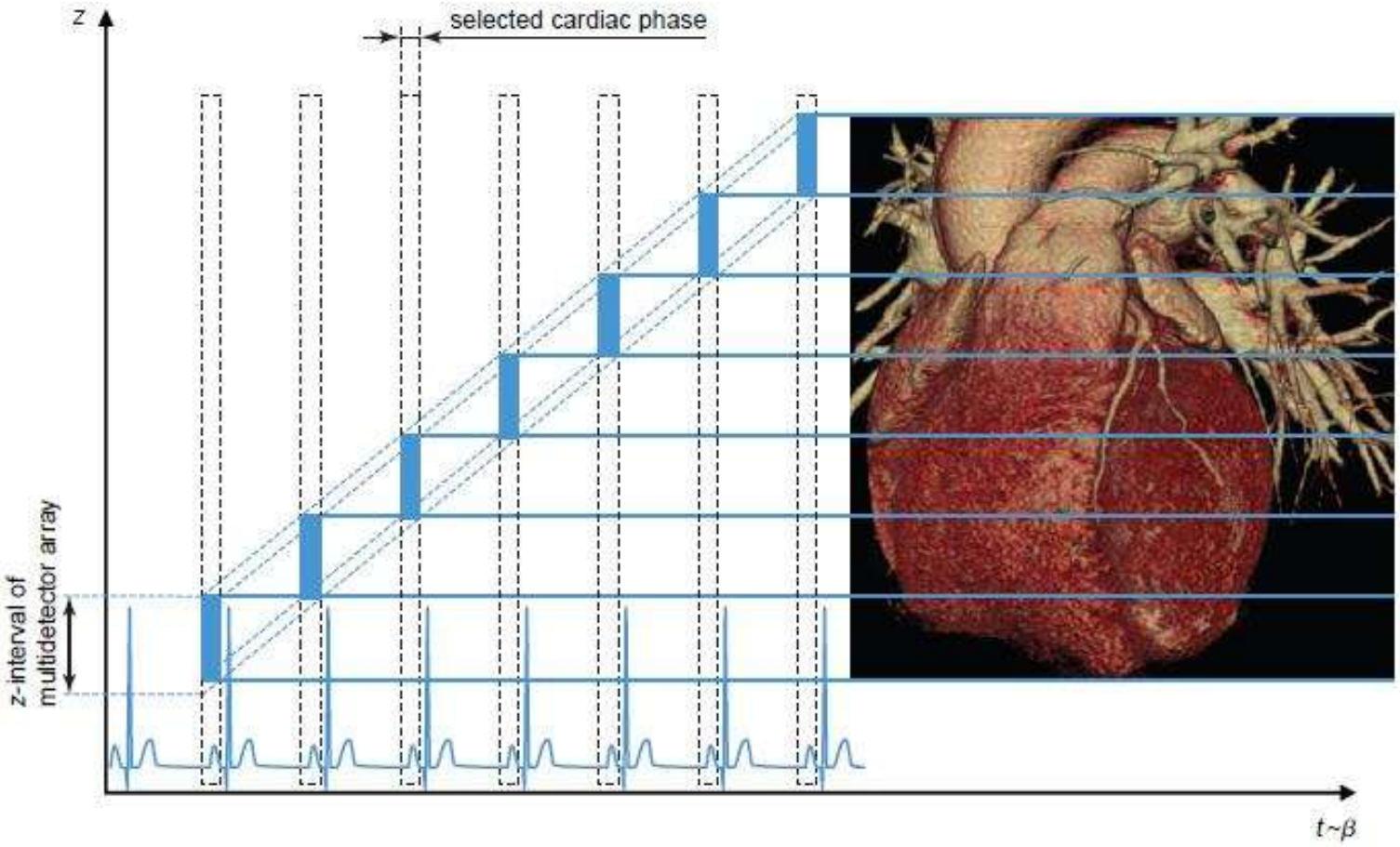
$$p(r,\theta) = \mathcal{R}\{f(x,y)\} = \int_{-\infty}^{\infty} f(r\cdot\cos\theta - s\cdot\sin\theta, r\cdot\sin\theta + s\cdot\cos\theta)\,\mathrm{d}s.$$

Obtém-se o *sinograma* da imagem
- Sua *projeção* – transformada de Radon

Recuperação da imagem: *retroprojeção*
- Transformada de Radon inversa $f(x,y) = R^{-1}\{p(r,\theta)\}$

Introdução e Fundamentos

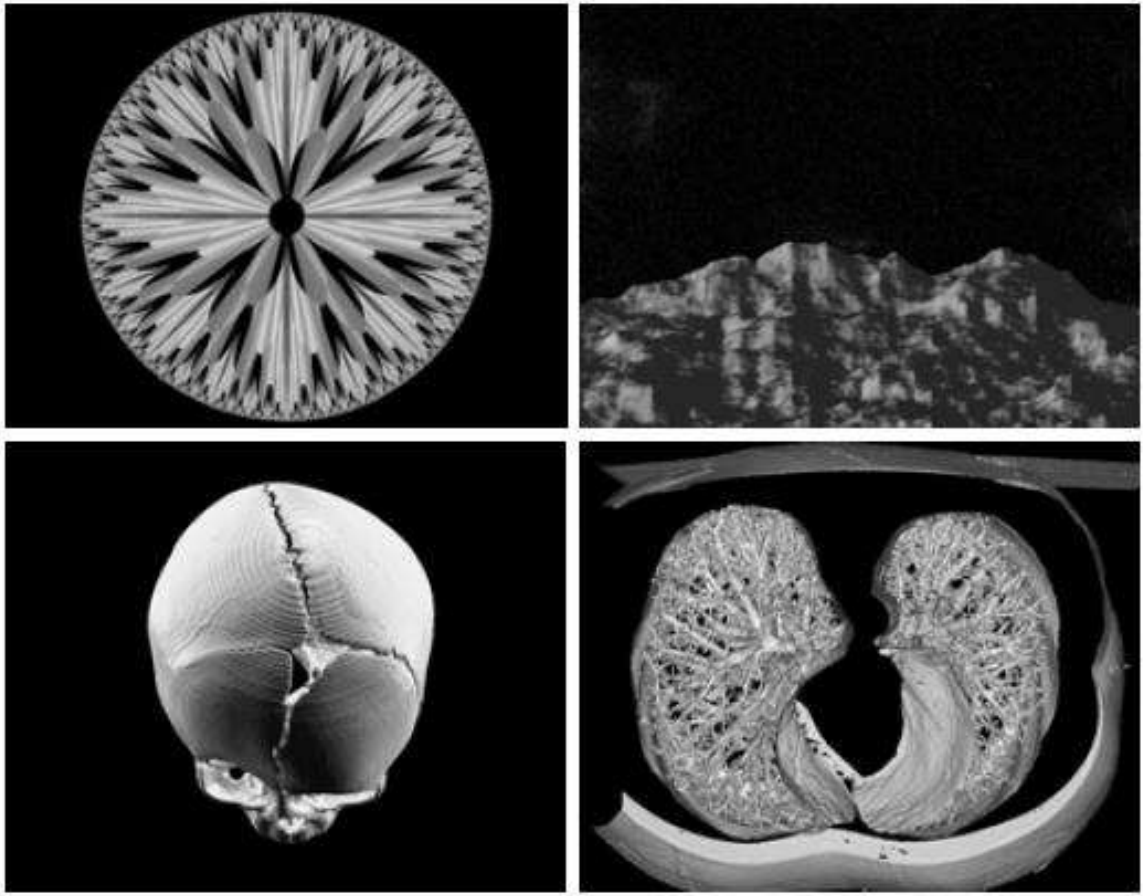
- Aplicações médicas – CT (imageamento cardíaco)



Principle of data acquisition for helical cardiac CT. The oblique lines represent the *z*-interval of the multidetector array as a function of time or angular tube position β.

Introdução e Fundamentos

- Imagens geradas digitalmente



a b
c d

FIGURE 1.22
(a) and (b) Fractal images. (c) and (d) Images generated from 3-D computer models of the objects shown. (Figures (a) and (b) courtesy of Ms. Melissa D. Binde, Swarthmore College, (c) and (d) courtesy of NASA.)

Introdução e Fundamentos

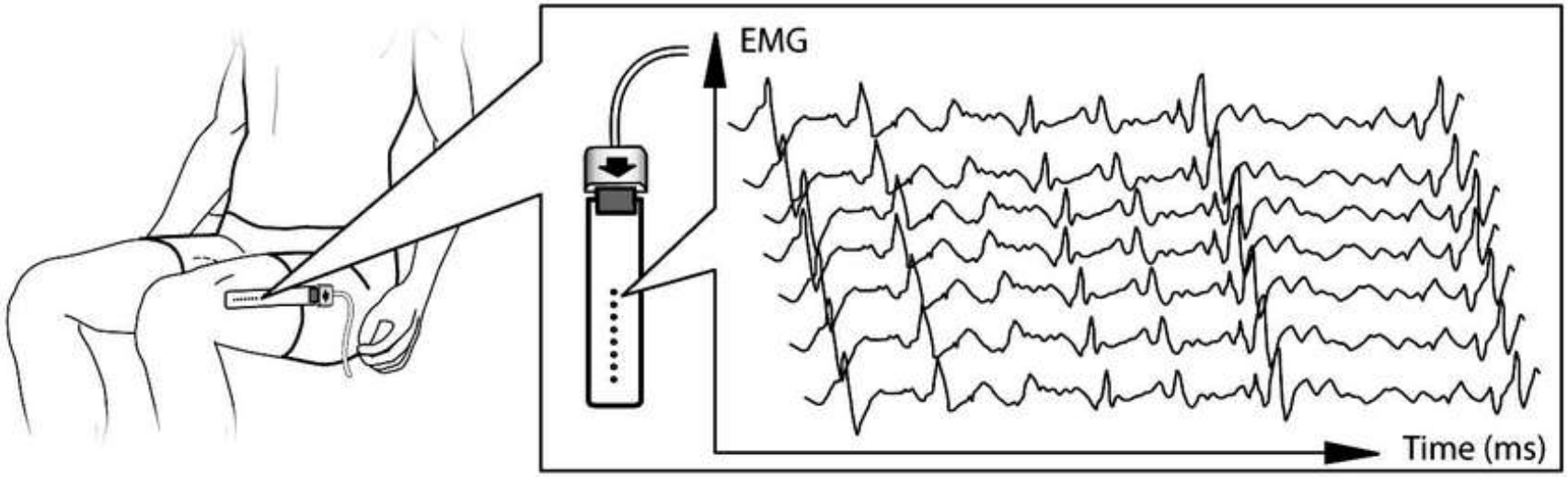
- Informações que originalmente não constituem imagens

Introdução e Fundamentos

- Informações que originalmente não constituem imagens
  - Recuperação de sinais gravados em mídias físicas

Introdução e Fundamentos

- Informações que originalmente não constituem imagens
  - Recuperação de sinais gravados em mídias físicas
  - Estimação de velocidade de propagação de sinais eletrofisiológicos (detecção multicanal)



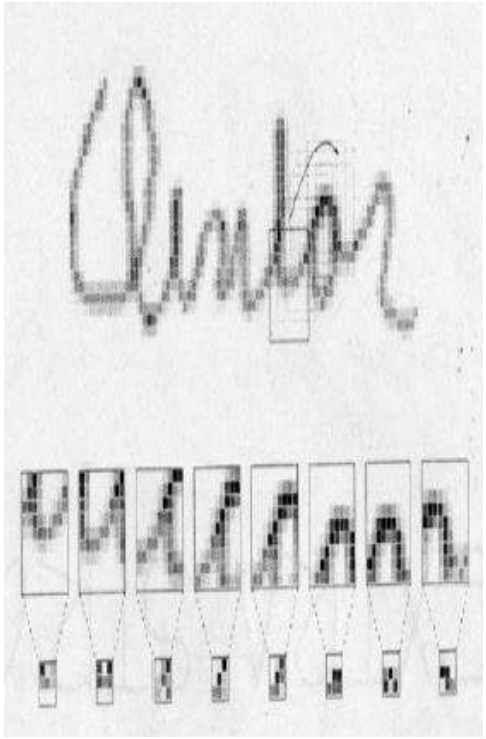
Introdução e Fundamentos

- Detecção e reconhecimento facial



Introdução e Fundamentos

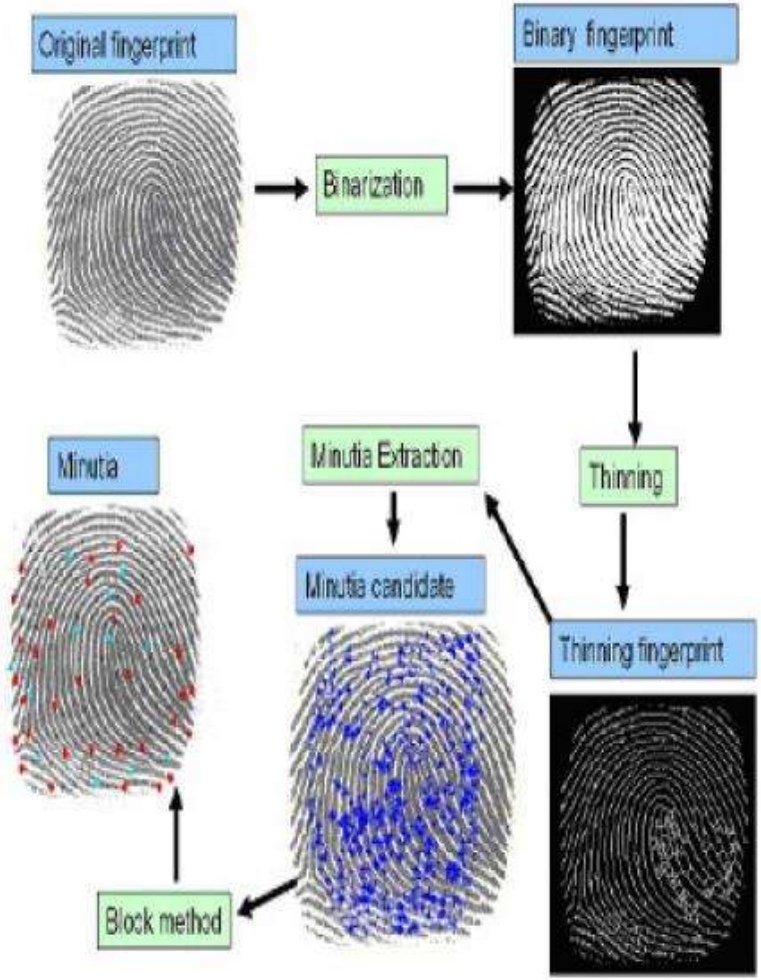
- Reconhecimento de grafia





Introdução e Fundamentos

- Biometria – impressão digital



Introdução e Fundamentos

- Biometria – íris

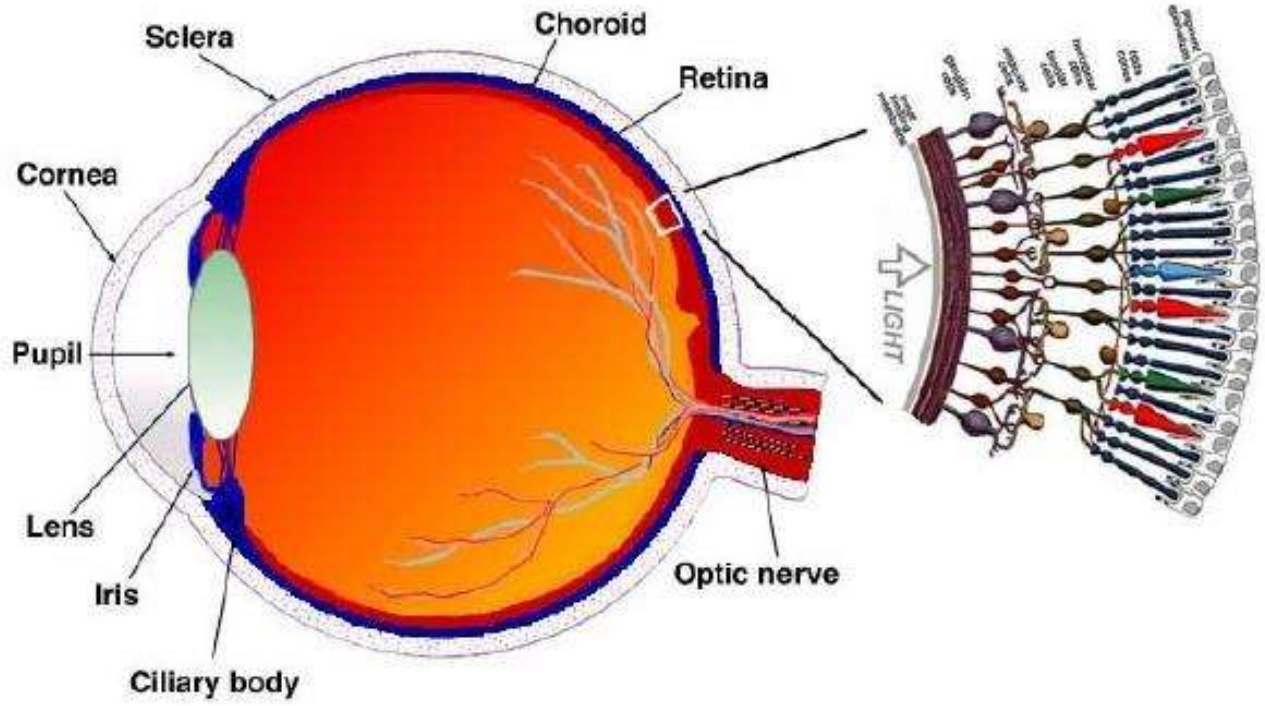


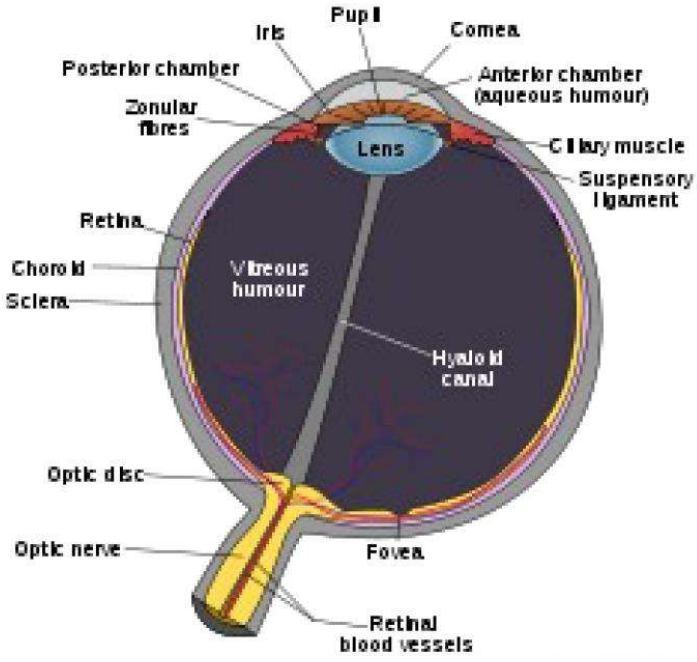
Introdução e Fundamentos

Sistema Visual Humano

Introdução e Fundamentos

- Sistema Visual Humano

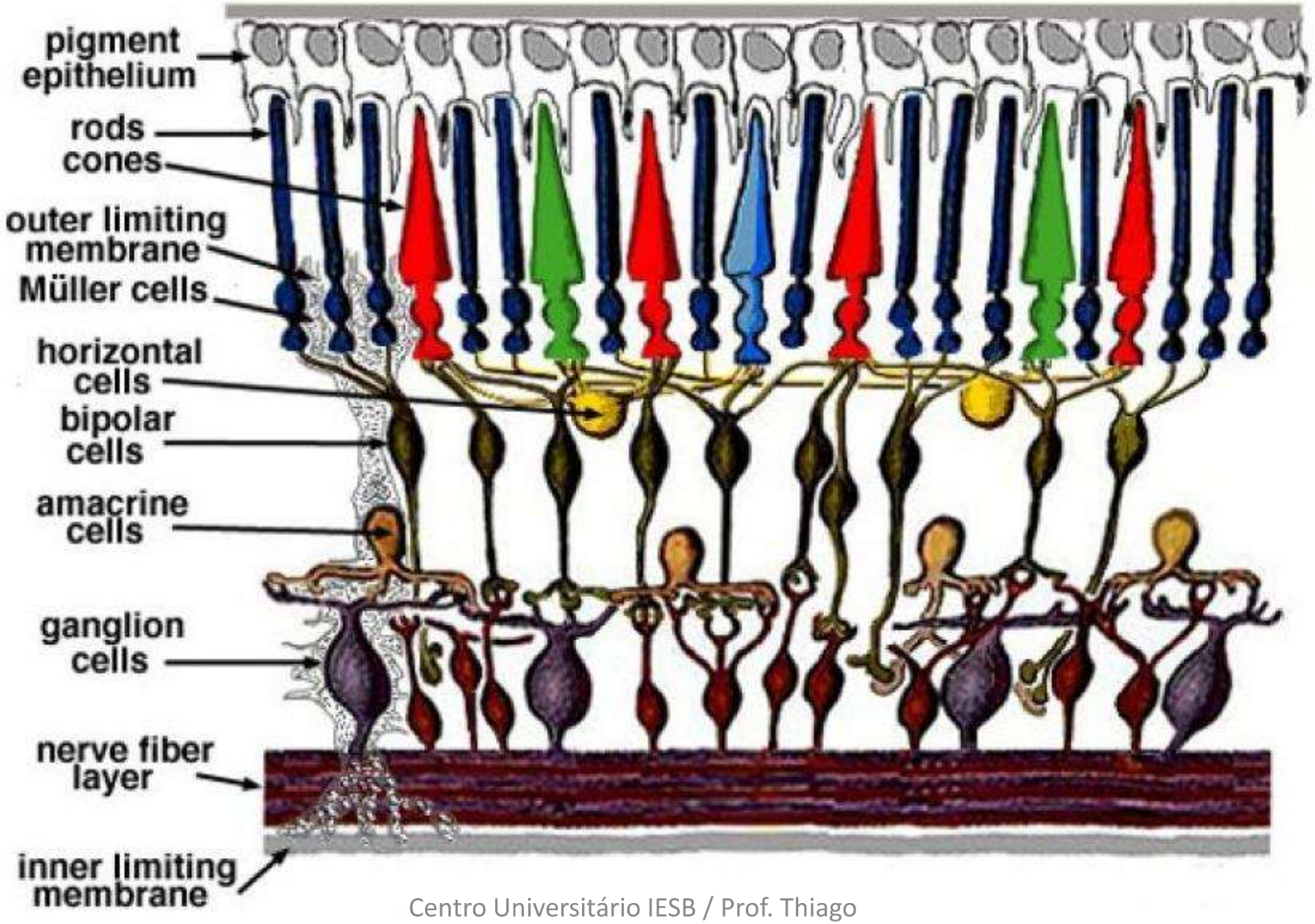




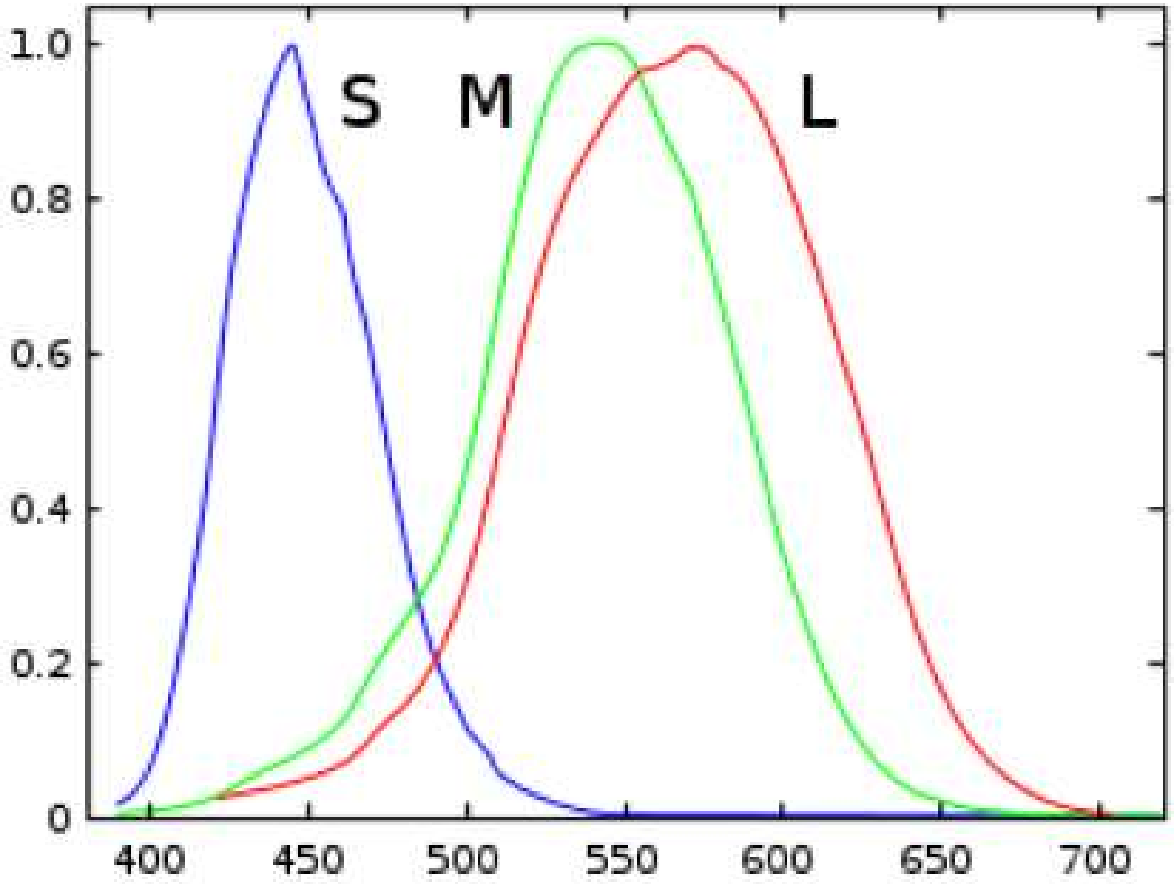
Introdução e Fundamentos

- Cones
  - (S,M,L)
  - ~ 5 milhões
  - Percepção luminosa
  - Intensidade luminosa média
  - Centro da retina
  - Detalhes da imagem
- Bastonetes
  - Intensidade luminosa baixa
  - ~ 100 milhões
  - Distribuídos ao longo da retina

Introdução e Fundamentos

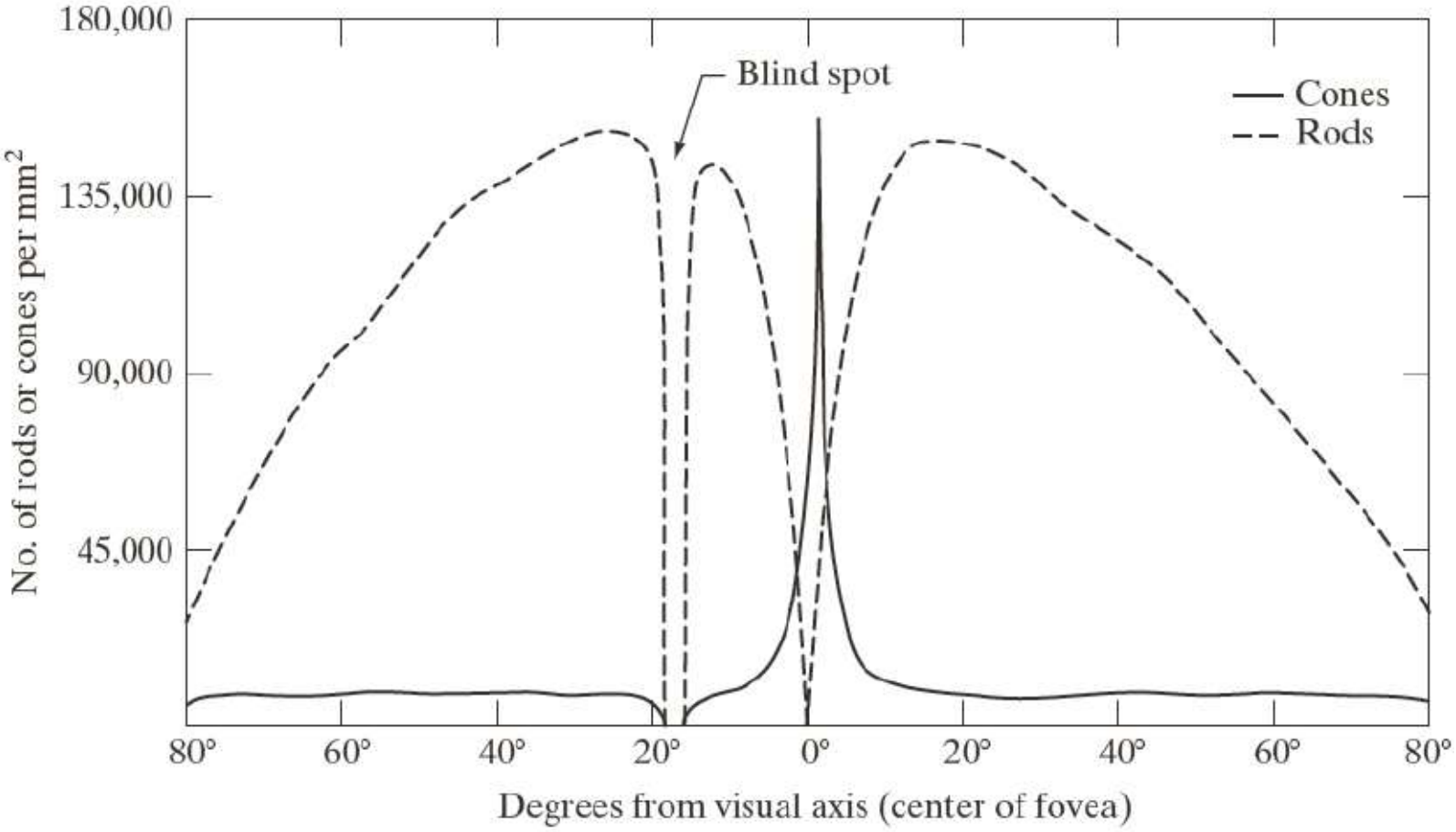


Introdução e Fundamentos



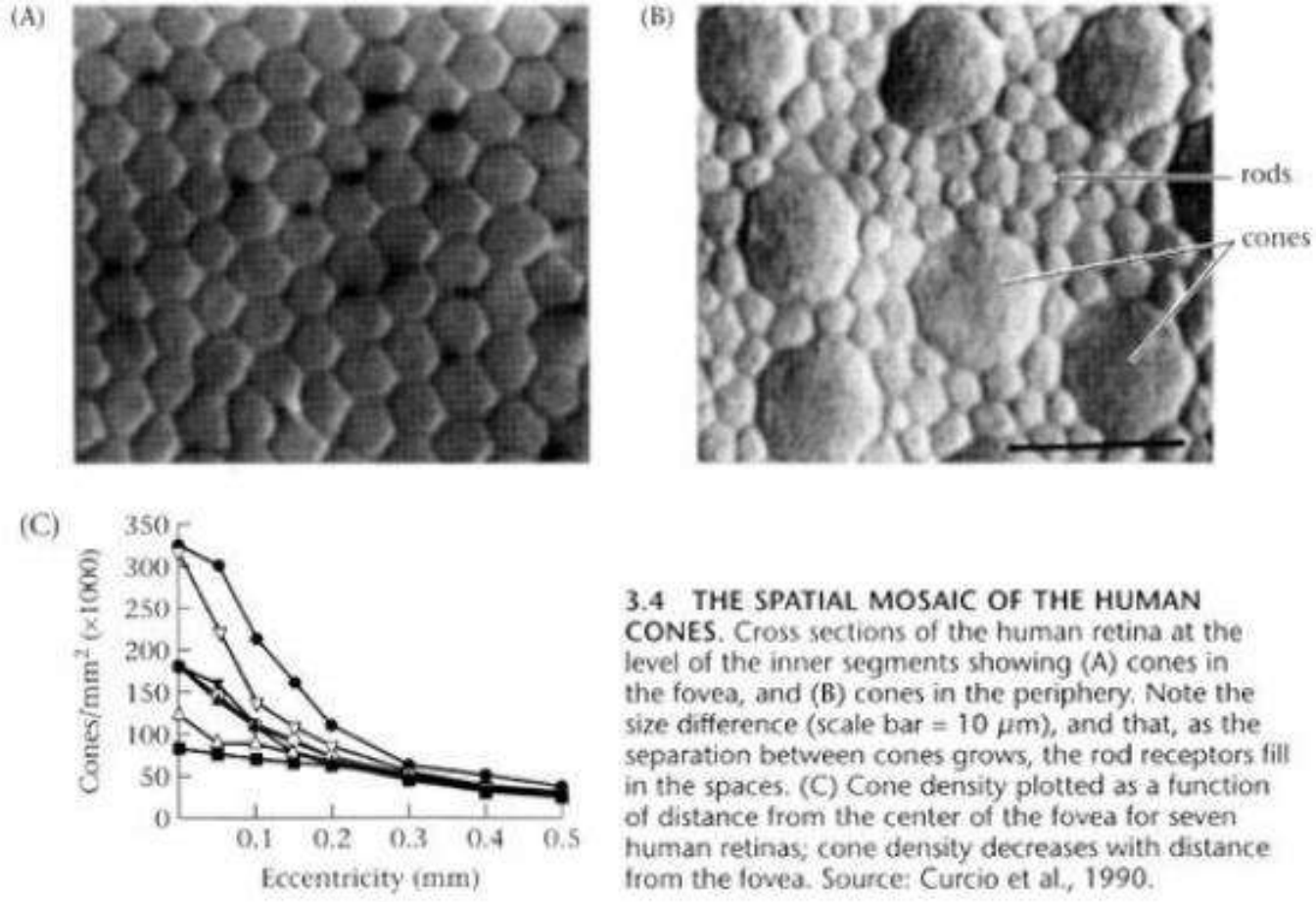
Introdução e Fundamentos

- Distribuição de cones e bastonetes
  - Densidade ao longo da retina



Introdução e Fundamentos

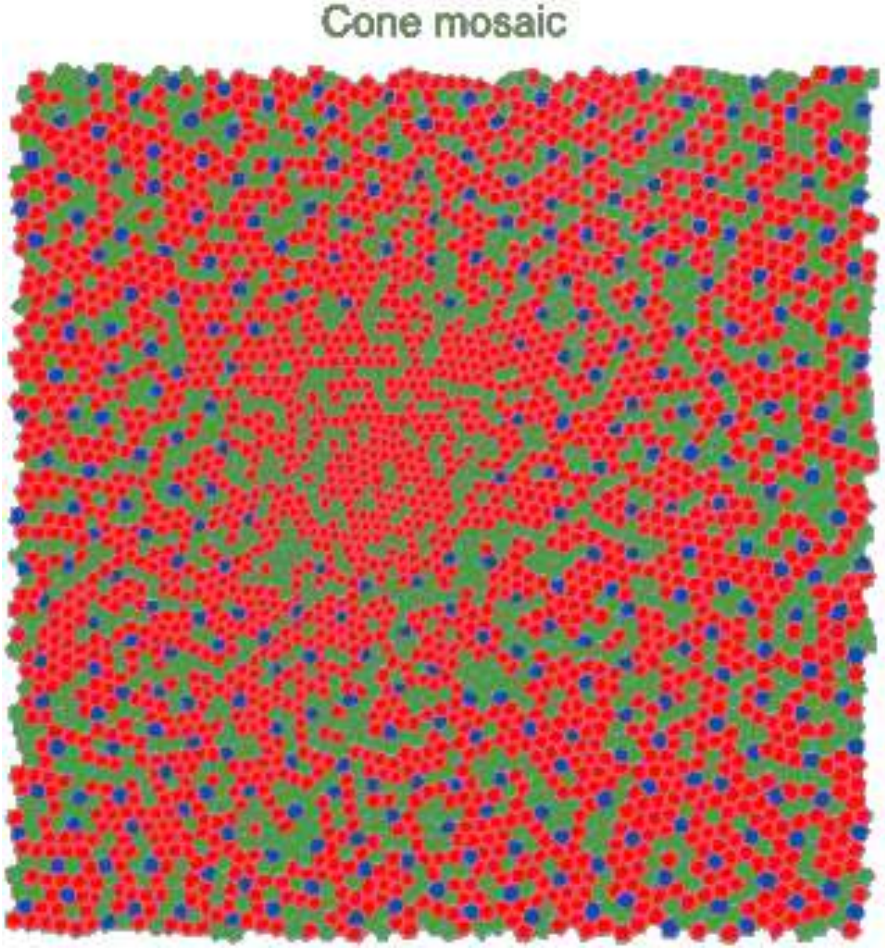
- Distribuição de cones e bastonetes
  - Densidade ao longo da retina



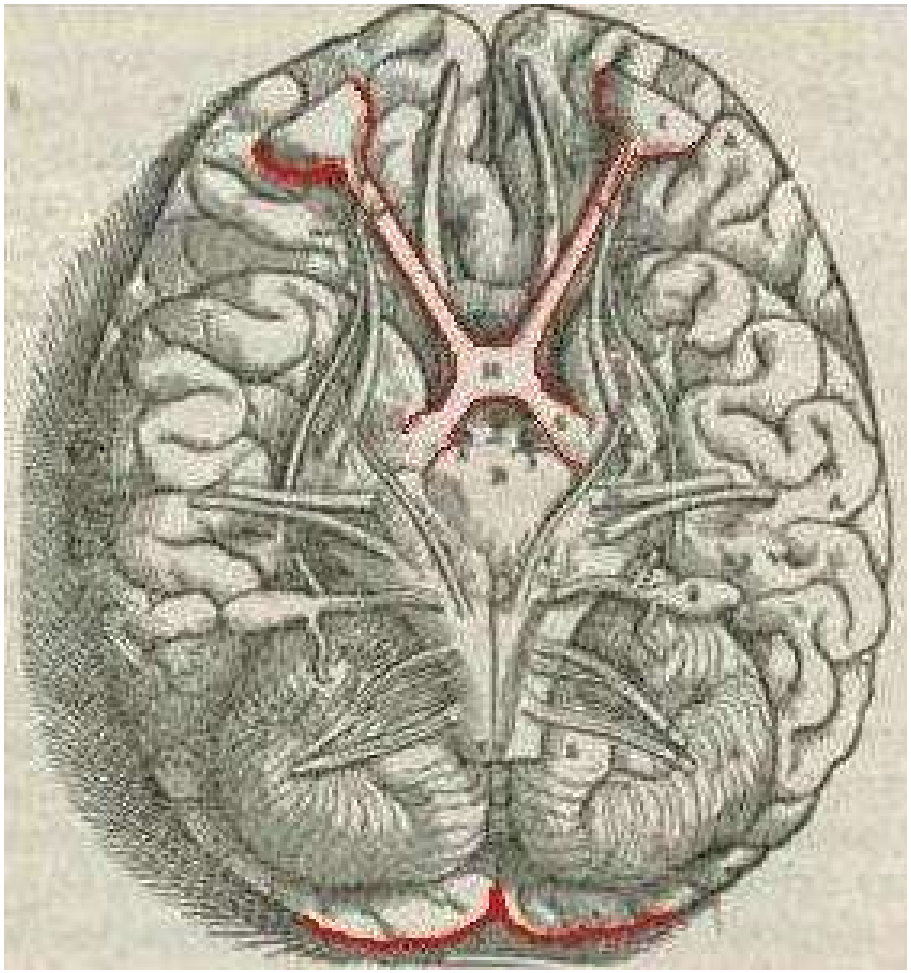
3.4 THE SPATIAL MOSAIC OF THE HUMAN CONES. Cross sections of the human retina at the level of the inner segments showing (A) cones in the fovea, and (B) cones in the periphery. Note the size difference (scale bar = 10 μm), and that, as the separation between cones grows, the rod receptors fill in the spaces. (C) Cone density plotted as a function of distance from the center of the fovea for seven human retinas; cone density decreases with distance from the fovea. Source: Curcio et al., 1990.

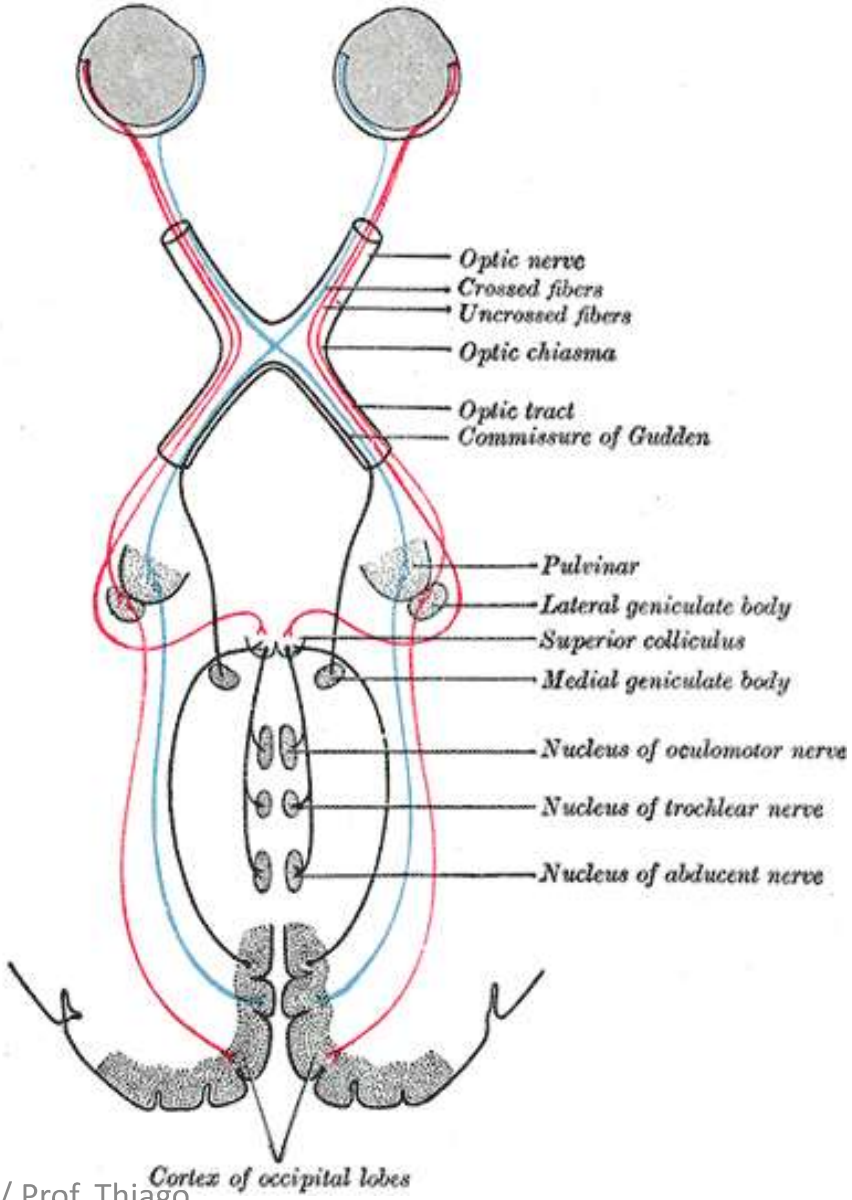
Introdução e Fundamentos

- Distribuição – fotorreceptores

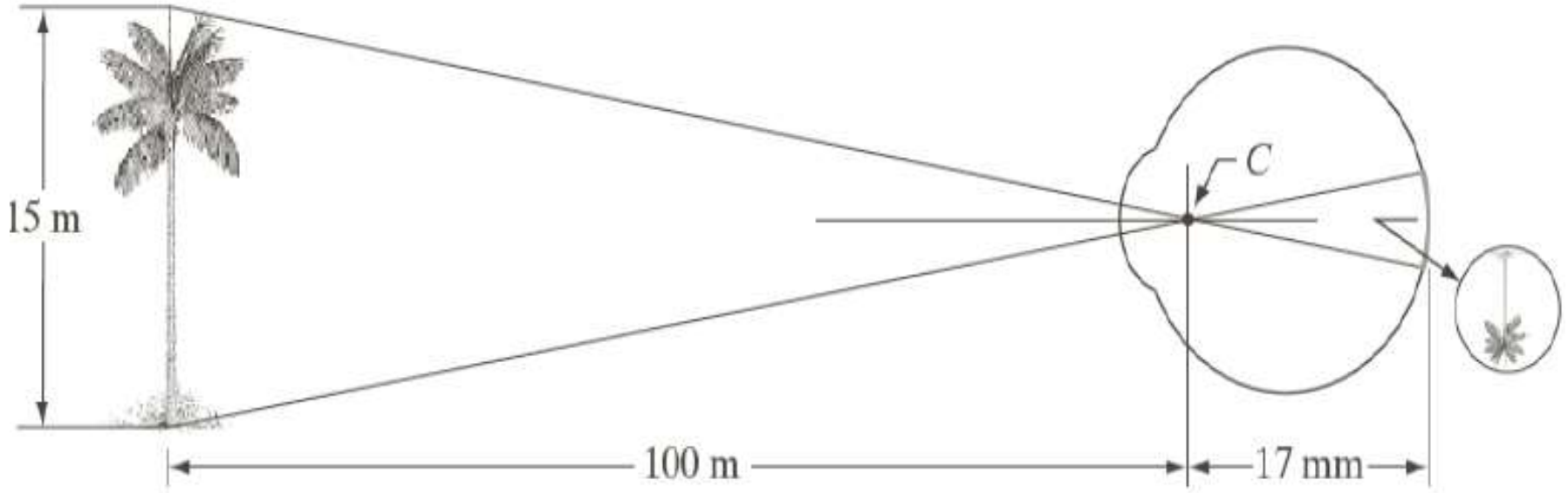


Introdução e Fundamentos





Introdução e Fundamentos



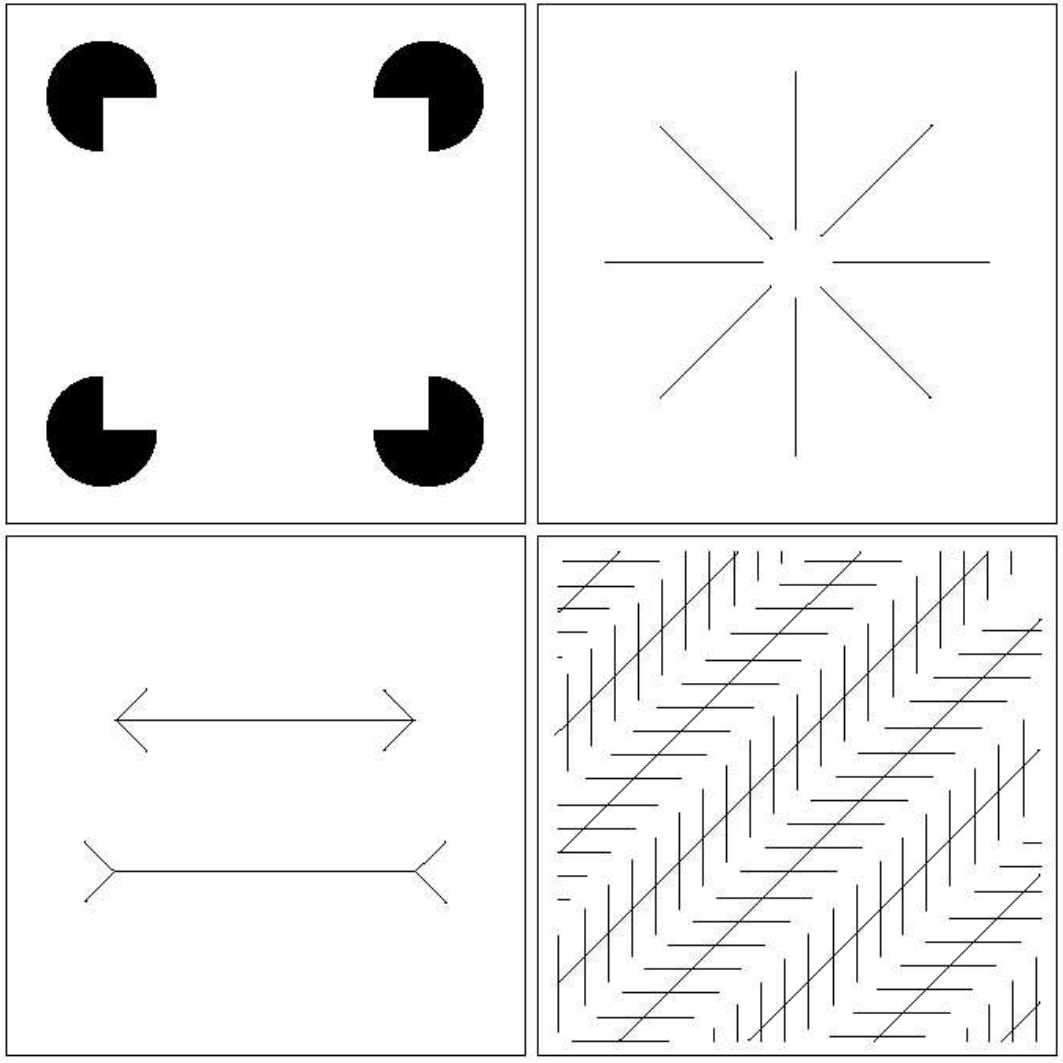
Graphical representation of the eye looking at a palm tree. Point *C* is the optical center of the lens.

Introdução e Fundamentos

Percepção – sistema visual humano

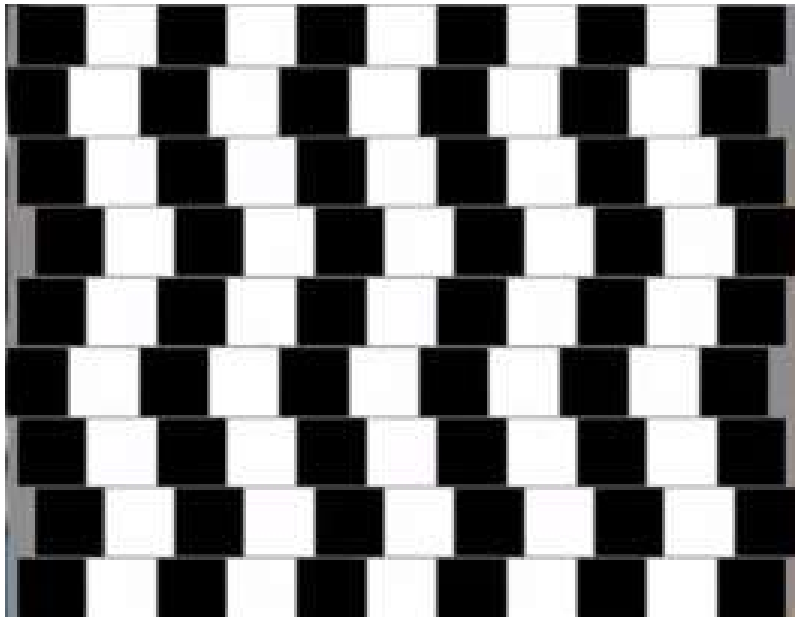
a b
c d

FIGURE 2.9 Some well-known optical illusions.

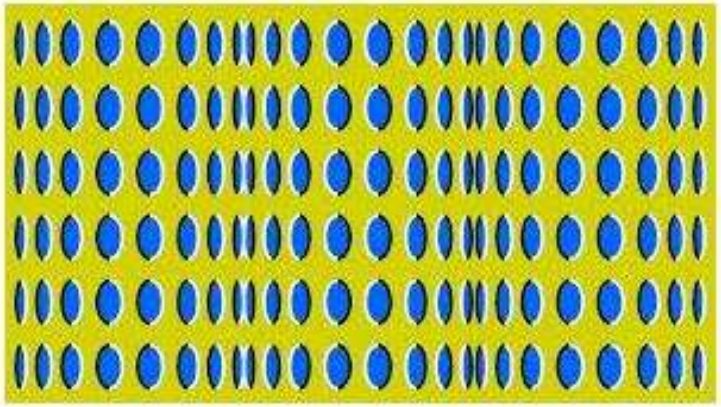


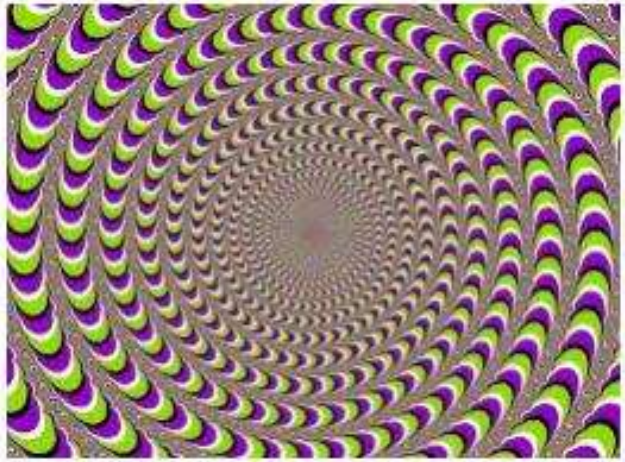
Introdução e Fundamentos

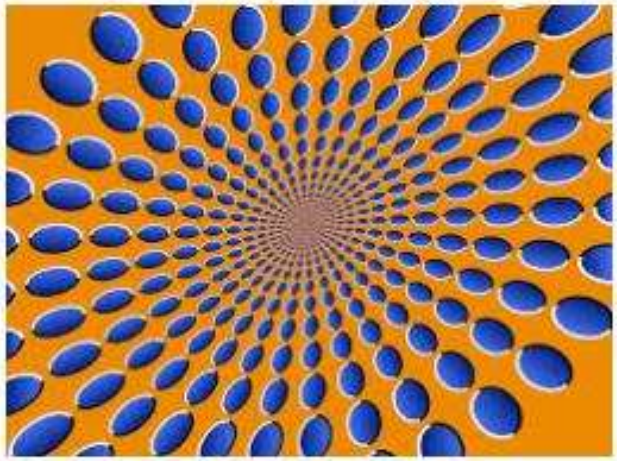
Percepção – sistema visual humano

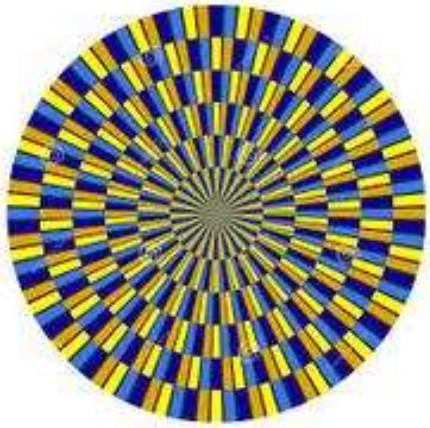


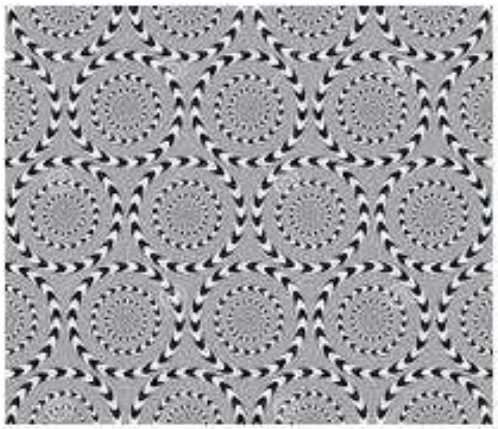






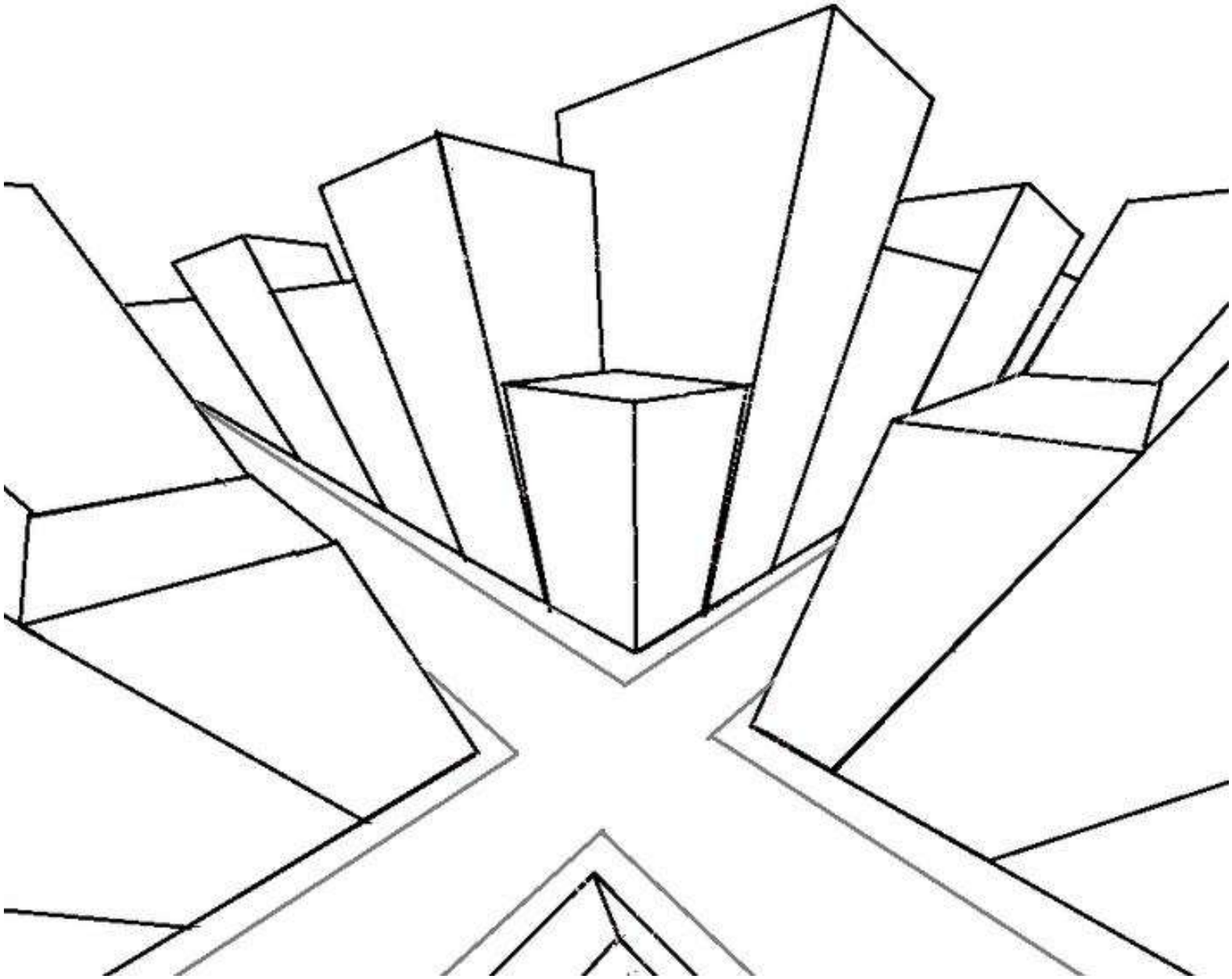












Introdução e Fundamentos

- Percepção – intensidade luminosa

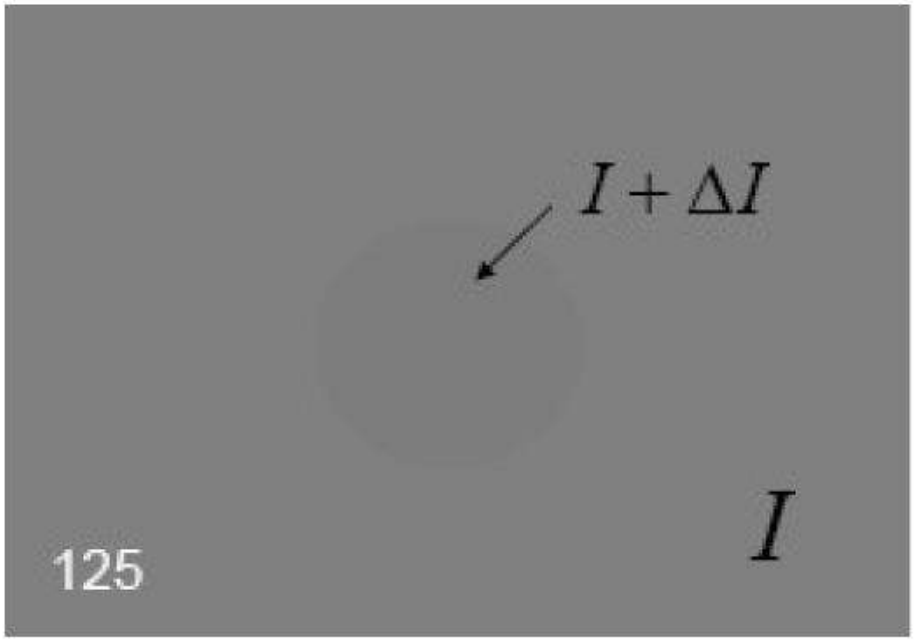


FIGURE 2.5 Basic experimental setup used to characterize brightness discrimination.

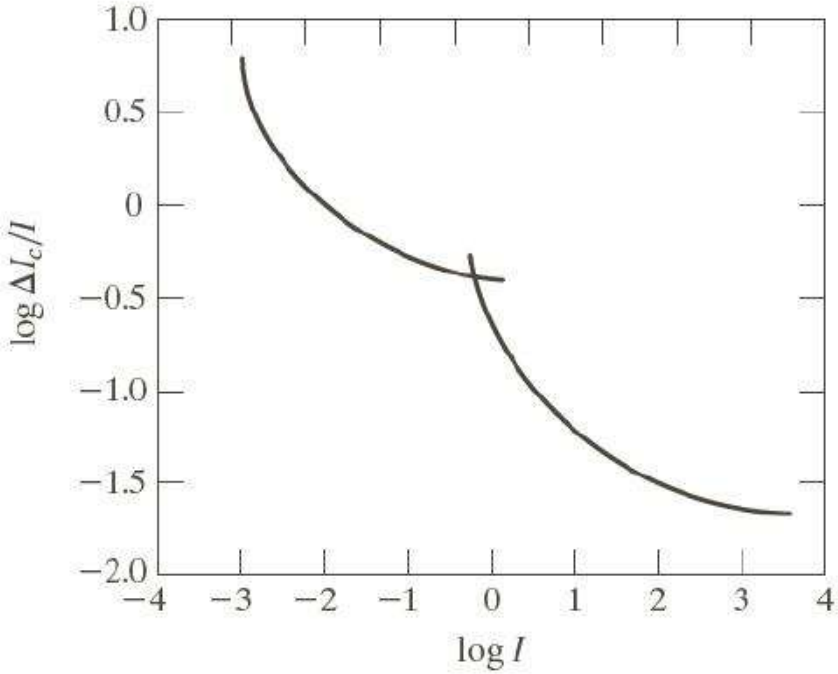
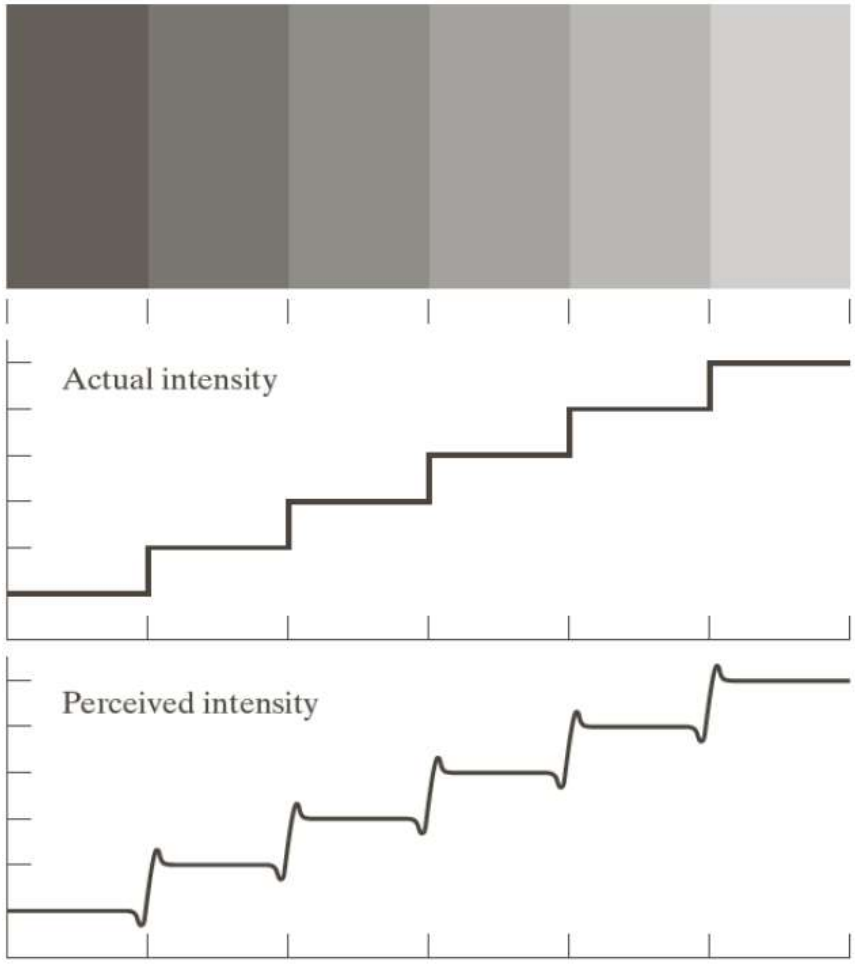


FIGURE 2.6 Typical Weber ratio as a function of intensity.

Introdução e Fundamentos

- Percepção – intensidade luminosa

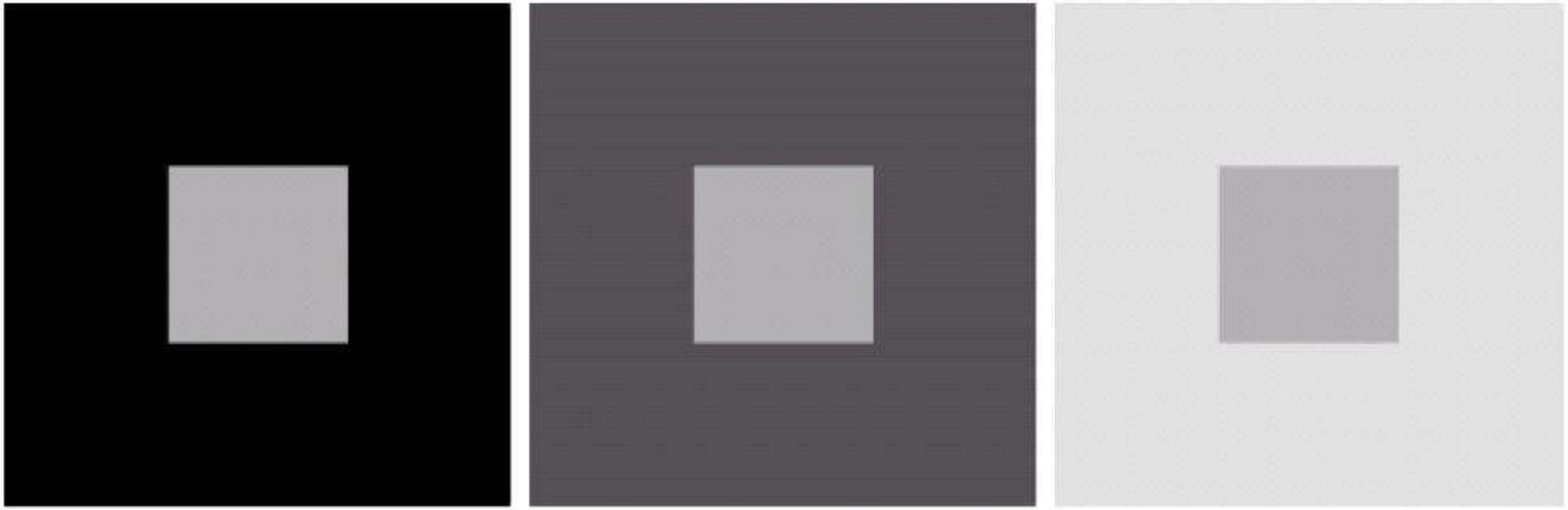


a
b
c

FIGURE 2.7
Illustration of the Mach band effect. Perceived intensity is not a simple function of actual intensity.

Introdução e Fundamentos

- Percepção – intensidade luminosa

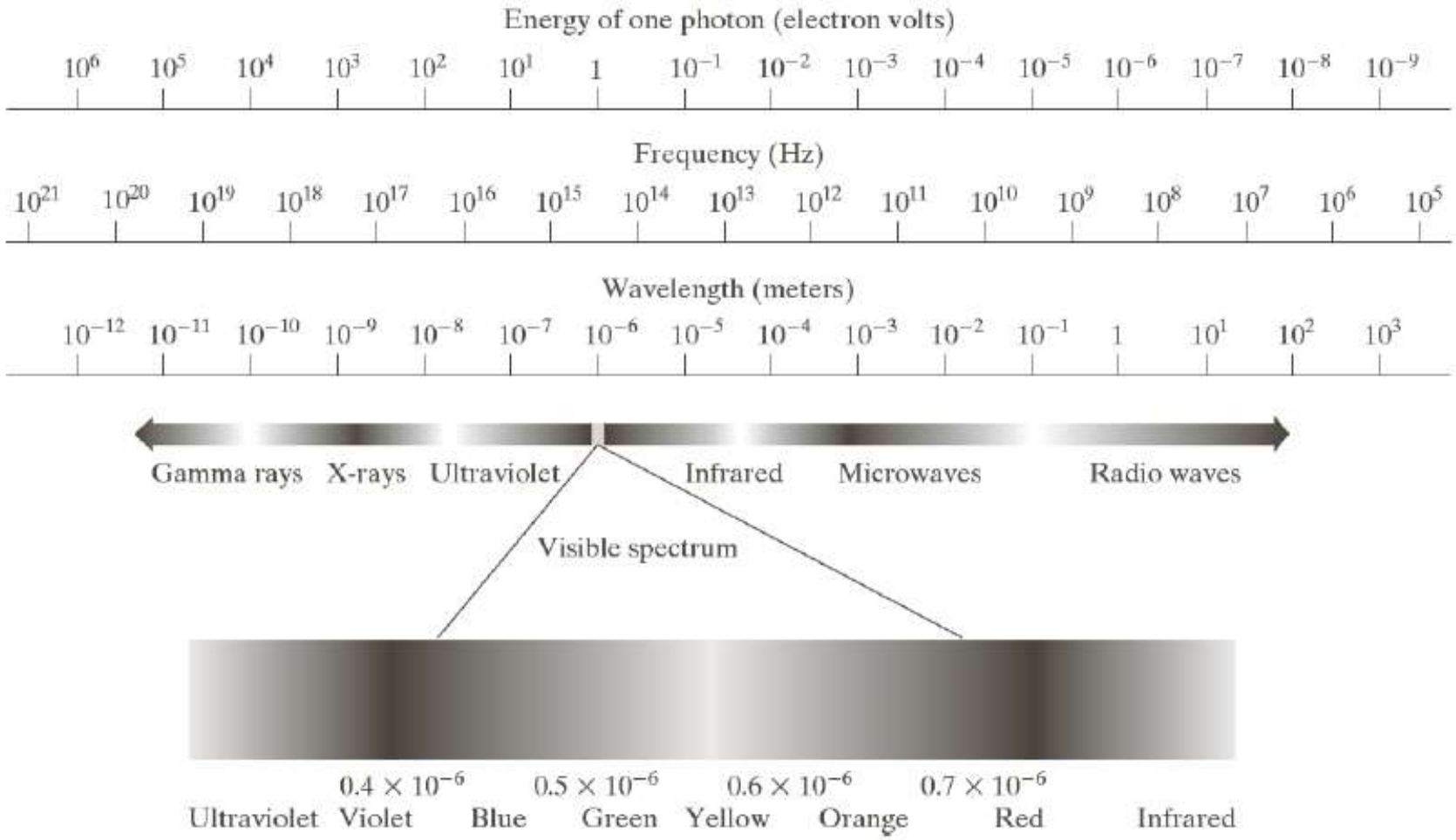


a b c

FIGURE 2.8 Examples of simultaneous contrast. All the inner squares have the same intensity, but they appear progressively darker as the background becomes lighter.

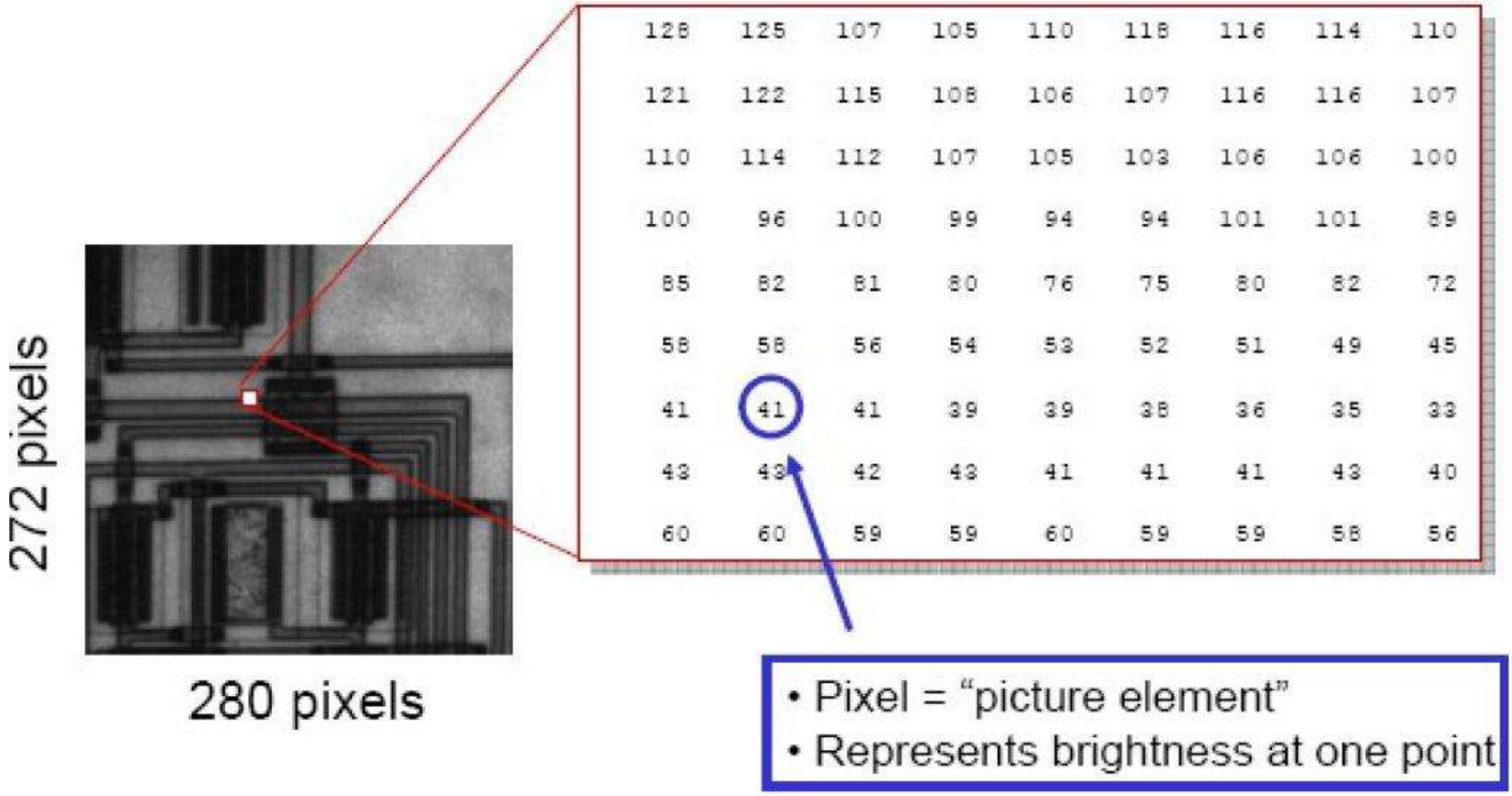
Introdução e Fundamentos

- Espectro Eletromagnético



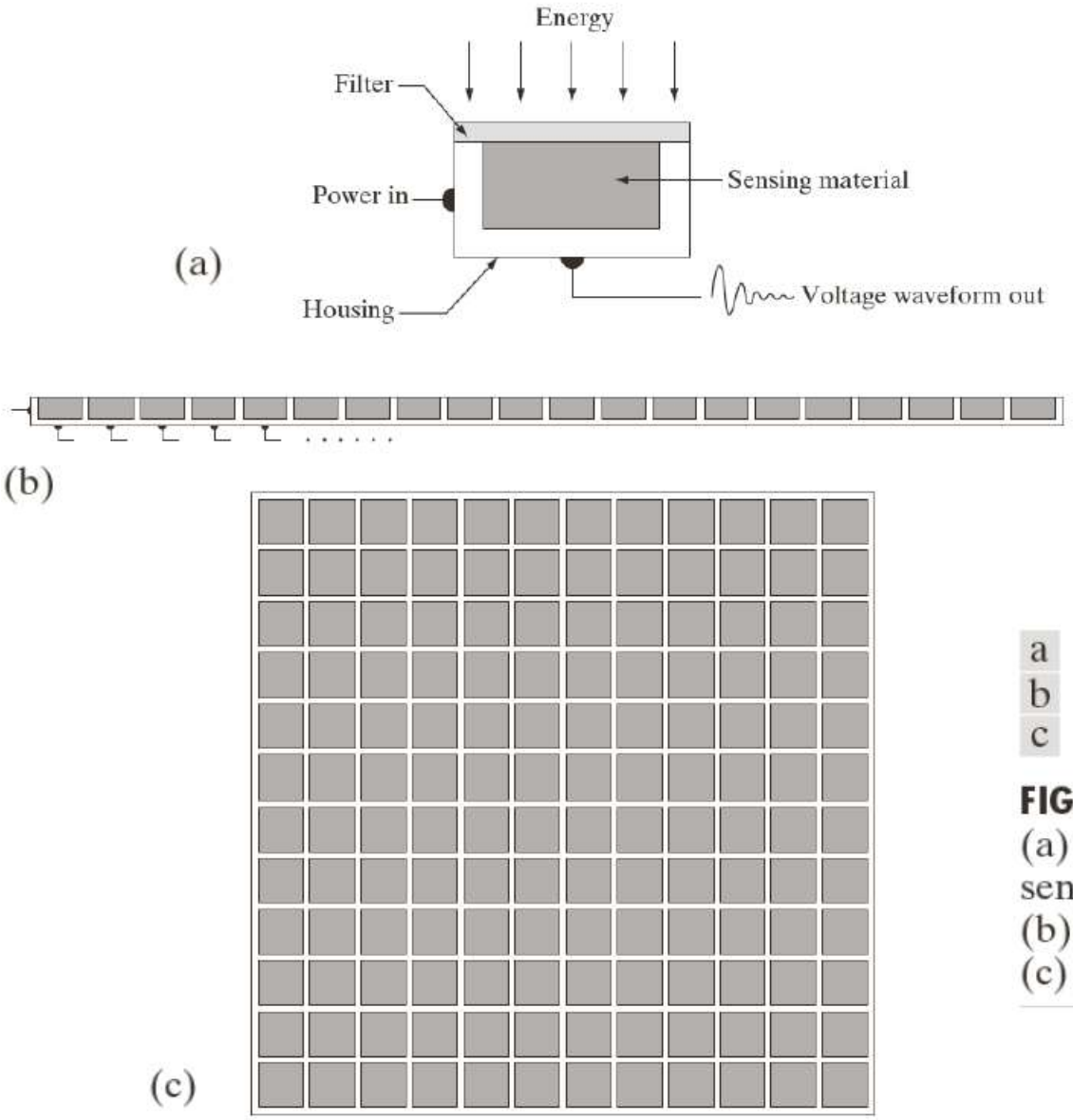
Introdução e Fundamentos

- Representação de uma imagem na forma digital



Introdução e Fundamentos

- Aquisição

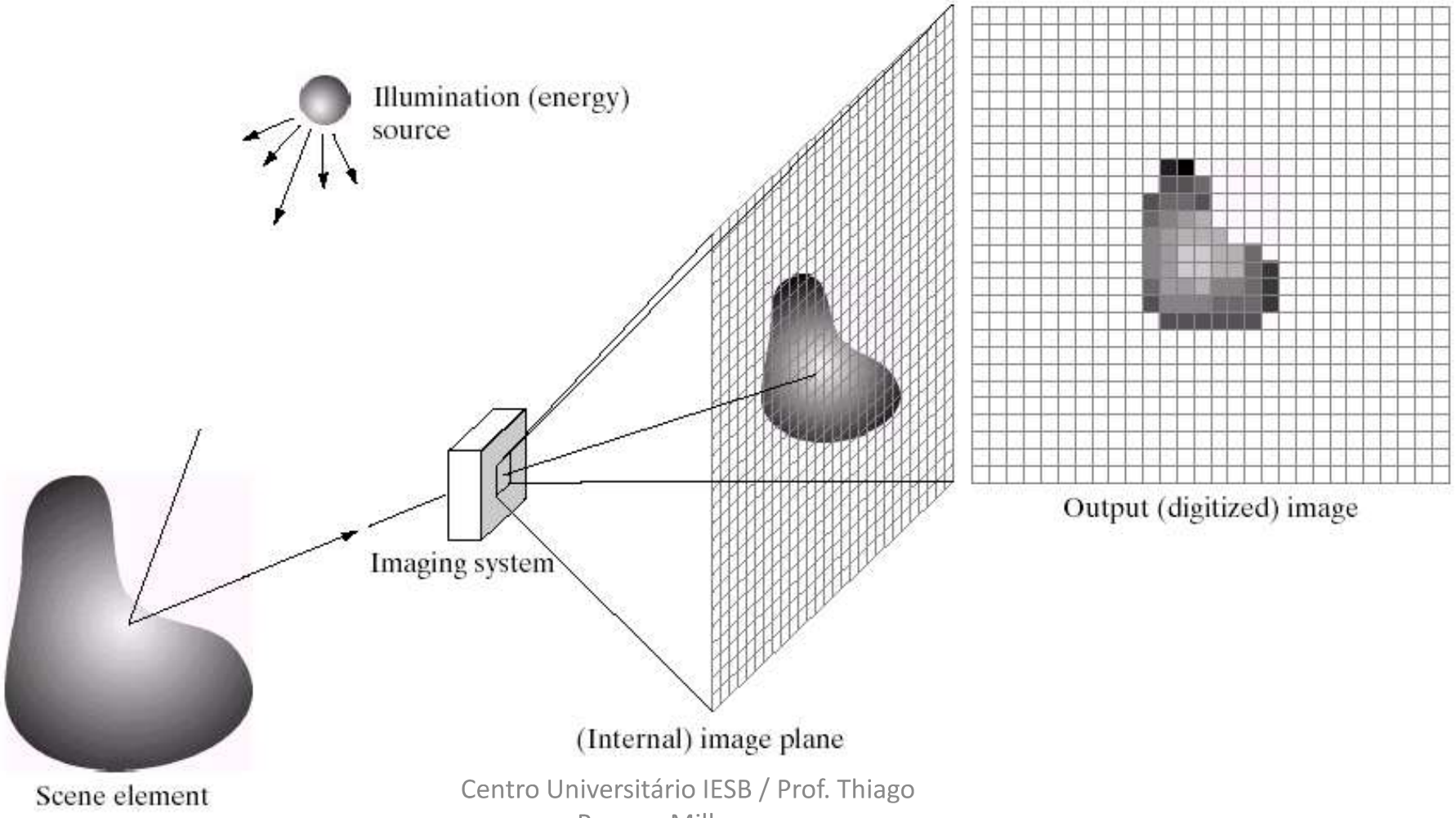


a
b
c

FIGURE 2.12
(a) Single imaging sensor.
(b) Line sensor.
(c) Array sensor.

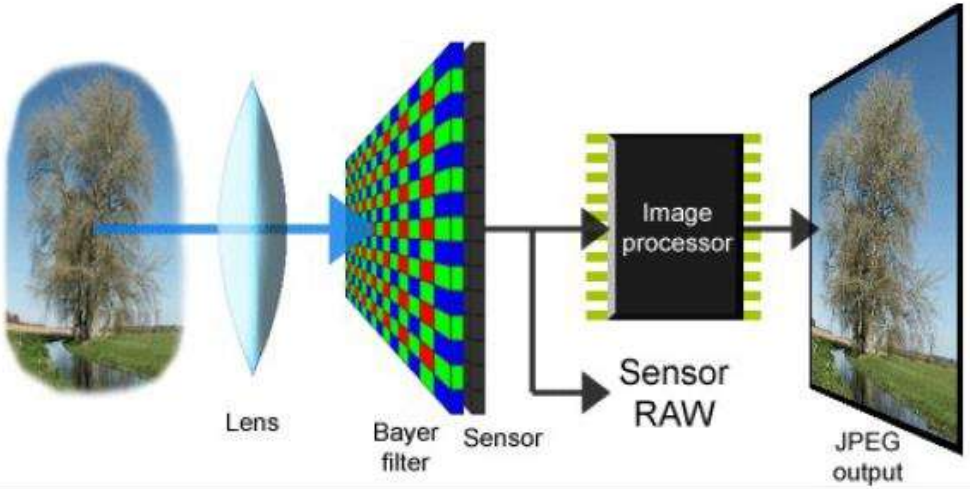
Introdução e Fundamentos

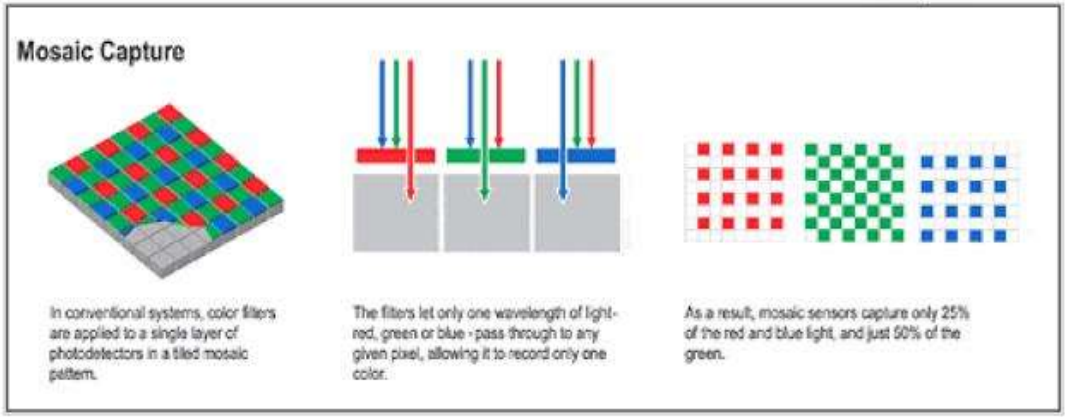
- Aquisição

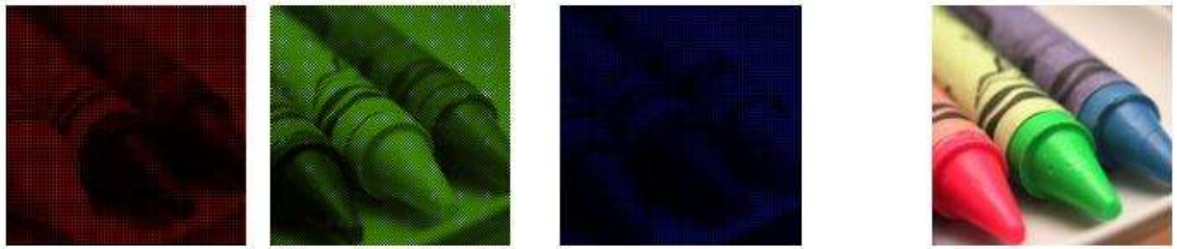


Introdução e Fundamentos

- Aquisição
  - Cores

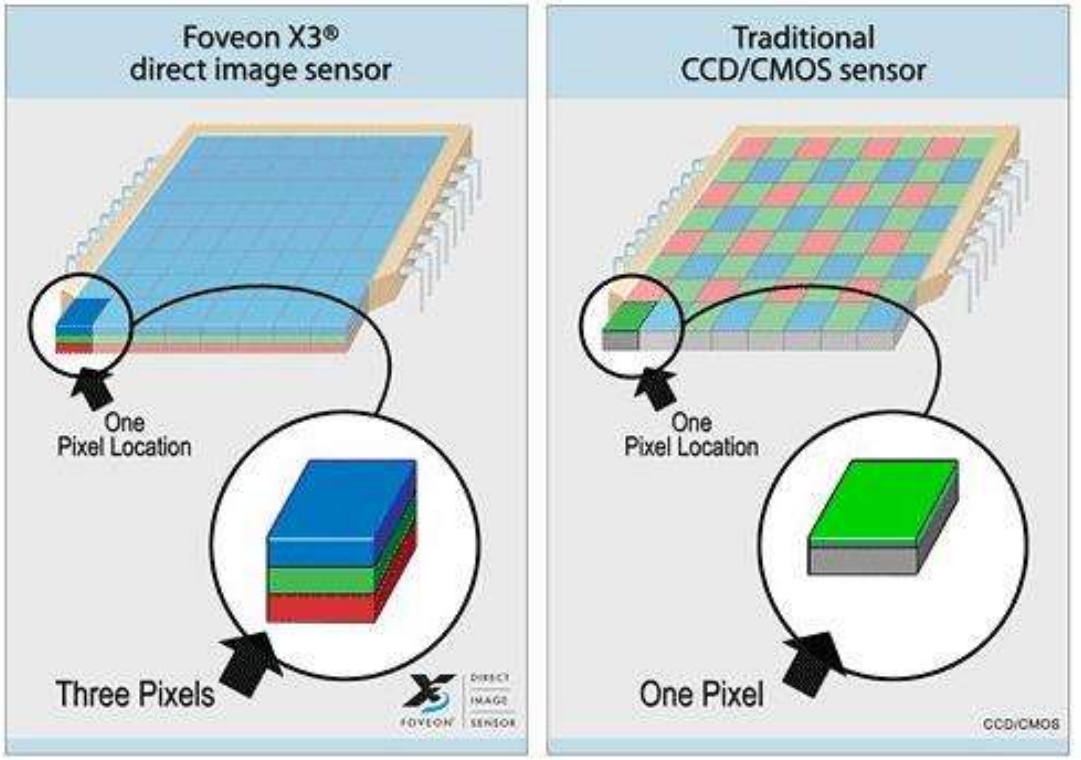






Introdução e Fundamentos

- Foveon X3



Introdução e Fundamentos

- Comparativo

Filtro em mosaico | Filtro Foveon

